Q1.

Find any relative extrema of
$f(x) = x^2\sqrt{16 - x^2}$

State where the function is increasing or decreasing and determine points of inflection.

Q2.

Find the absolute maximum and minimum values of
$f(x) = x^2 + \frac{64}{x}$

on $(0, +\infty)$.

Q3.

The total cost and total revenue functions are
$C(x) = \frac{1}{2}x^3 - 10x^2 + 60x + 500$
$R(x) = 150x - 2x^2$

a) Find the profit function $P(x)$ and determine the level of output for maximum profit.

b) Find $MR(x)$ and $MC(x)$.

Q4.

A cylindrical can is to have a volume of $1500\ \text{cm}^3$.

Find the dimensions (radius and height) that minimize the surface area.